Technical Report

Time-Aware LightGBM Pipeline for ASHRAE GEPIII Building Energy Prediction with SHAP and Meter-Wise Error Analysis

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This project was developed as part of the Applied Machine Learning module within the MSc Data Science and Analytics programme at Cardiff University, United Kingdom.

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1. Introduction

Building energy consumption is difficult to predict because it depends on interacting physical, environmental, temporal, and operational factors. Building size, functional use, meter type, weather conditions, site characteristics, and seasonal usage patterns can all influence consumption. This project challenged this problem under the ASHRAE Great Energy Predictor III (GEPIII) framework, where the aim is to predict building-level energy consumption from timestamped meter readings, building metadata, and site-level weather observations. The task is formulated as a supervised regression problem using the log-transformed meter reading as the modelling target. The main focus of this project is to develop a compact and analytically justified machine learning pipeline for building energy prediction. The pipeline combines data integration, missing-value handling, domain-specific target correction, logarithmic transformation, temporal feature engineering, model-specific encoding, and gradient-boosted tree modelling. Rather than relying only on model complexity, the project investigates how preprocessing, feature representation, and identifier-dependent variables affect predictive performance. Linear Regression, Ridge Regression, Random Forest, and LightGBM are compared under a chronological evaluation design suitable for timestamped energy data. The preferred final model is a tuned LightGBM variant that excludes building_id to reduce reliance on building-specific identifiers, achieving a RMSE of 1.184, MAE of 0.690, and R^2 of 0.680 on the log-transformed target, where the RMSE on the log-transformed target is mathematically equivalent to the RMSLE used in the Kaggle competition. SHapley Additive exPlanations (SHAP) analysis and meter-wise error analysis are then used to examine feature importance, model behaviour, and category-specific prediction difficulty. Lastly, considerations were given to social implications and future works.

2. Related Work

Building energy consumption prediction is a well-established application of data-driven modelling because accurate forecasts can support energy management, demand planning, building performance evaluation, and efficiency improvement. Amasyali and El-Gohary's review shows that machine learning has been widely applied in this area, but also emphasises that performance depends on more than algorithm choice alone. Data quality, preprocessing, feature selection, temporal granularity, and evaluation design all affect predictive reliability [1], [2]. This is particularly important because building energy use is shaped by multiple interacting factors, including building characteristics, weather conditions, operational schedules, occupant behaviour, and meter type [3], [4]. The task is therefore not a simple static regression problem, but a heterogeneous prediction problem requiring careful data preparation and feature representation. The dataset used in this project is closely related to the Building Data Genome Project 2 and the ASHRAE Great Energy Predictor III competition. Miller et al. describe this dataset as a large-scale collection of hourly energy meter readings from non-residential buildings, combined with building metadata and weather data [5]. This structure directly motivates the preprocessing pipeline used in the present project. Since the raw data are distributed across meter readings, building metadata, and site-level weather observations, the first requirement is to integrate these sources into a unified modelling schema. Previous work also highlights the importance of data cleaning, transformation, and feature engineering in building energy prediction [1]. This supports the preprocessing decisions made in this project, including missing-value handling, removal of weak or highly incomplete variables, unit correction, logarithmic transformation of skewed variables, and most importantly, extraction of temporal predictors from timestamps.

In terms of modelling, the literature suggests that linear models are useful baselines but are unlikely to capture the full complexity of building energy consumption. Linear Regression provides a transparent reference point for assessing how much signal can be captured under simple additive assumptions. Ridge Regression extends this baseline by applying L2 regularisation, which can improve stability when predictors are correlated or when one-hot encoding increases feature dimensionality [6]. However, because energy consumption depends on non-linear interactions between building size, weather, meter type, use category, and time, more flexible models are often required.

Tree-based ensemble methods are therefore particularly relevant for this task. Random Forest provides a useful non-linear benchmark because it can capture threshold effects and interactions without requiring them to be manually specified [7]. Gradient boosting extends tree-based modelling by sequentially correcting prediction errors, making it especially effective for structured tabular datasets [8]. LightGBM builds on this family of methods and is designed for efficient learning on large, high-dimensional datasets [9], [10]. Prior work on the ASHRAE GEPIII competition further shows that top-performing solutions relied heavily on gradient-boosted tree ensembles, particularly LightGBM, and that preprocessing and feature extraction were key differentiators [11]. This supports the modelling strategy in the present project, where LightGBM is treated as the main model after comparison with linear and Random Forest baselines.

Evaluation design is another important issue in timestamped energy data. Random train-test splitting can produce overly optimistic results by allowing future patterns to influence training. Time-aware validation methods preserve chronological order by training on earlier observations and evaluating on later observations, which better reflects future-period prediction [12]. This justifies the chronological train-validation-test design used in this project. Finally, because high-performing ensemble models are less directly interpretable than linear models, SHAP (SHapley Additive exPlanations) was used to examine feature contributions [10], [13]. This allowed the final LightGBM model to be assessed not only by predictive accuracy, but also by whether it relied on meaningful predictors such as building size, site, meter type, weather, and temporal features [14]. Overall, the literature supports the project's main design: a compact but carefully engineered LightGBM pipeline with explicit preprocessing, time-aware feature engineering, chronological evaluation, and post-hoc interpretability analysis.

3. Problem Setting and Dataset

This project uses the Kaggle's[1] ASHRAE GEPIII Competition dataset, consisting of meter readings, building metadata, and weather observations. The modelling table was formed from more than 20 million timestamped records across 1,449 buildings, four meter types, and 16 primary-use categories. The target was meter reading, later corrected and transformed using log-plus-one. Since the data are time-indexed, the task was treated as future-period prediction rather than random static regression: earlier months were used for training and validation, and later months were reserved for testing. All exploratory analysis, preprocessing, feature engineering, modelling, evaluation, and visualisation were implemented in Python using Pandas, NumPy, Matplotlib, scikit-learn, LightGBM, and SHAP.

4. Exploratory Data Analysis (EDA)

EDA was conducted to understand the structure, quality, and modelling implications of the dataset before feature engineering and model training. Since the task involved timestamped building-level meter readings combined with building metadata and site-level weather observations, the EDA focused on four main aspects: missing data, target distribution, building and meter heterogeneity, and temporal consumption patterns (Table 1). Findings were used to directly inform subsequent preprocessing and feature engineering decisions.

Table 1. Summary table of key Exploratory Data Analysis implications for modelling.

EDA finding	Modelling implication
Target variable was extremely right-skewed	Used log-transformed meter reading as the target
Building size was right-skewed	Used square feet log
Several metadata/weather variables had high missingness	Removed floor count, year built, cloud coverage, etc.
Energy consumption varied by month and meter type	Used time-aware features and chronological split
Meter types had different distributions	Kept meter as a categorical feature and reported meter-wise performance
Weather variables had low missingness and domain relevance	Retained air temperature, dew temperature, and wind speed
Repeated building-level observations may cause identifier dependence	Tested LightGBM with and without building id

4.1. Missing values and data quality

The missing-value analysis showed that several metadata and weather variables had substantial missingness. In particular, floor count, year built, and cloud coverage had high missing-value percentages, making them risky to impute reliably (Table 2). In contrast, air temperature, dew temperature, and wind speed had very low missingness and were therefore retained for modelling. This exploratory data analysis (EDA) finding directly informed the preprocessing strategy: highly incomplete variables were removed, while low-missingness continuous weather variables were

[1] Kaggle is a data science competition platform and online community for data scientists and machine learning practitioners under Google LLC. Website: https://www.kaggle.com

median-imputed (Section 5.2). The timestamp alignments including daylight saving shifts were identified as data quality concerns. Weather observations are recorded in UTC while meter readings follow local time. However, these offsets were not corrected in this project. Since the UTC offset is constant per site, human activity patterns remain broadly aligned with the raw timestamps [5].

Table 2. Missing-value summary.

Features	Missing percentage (%)	Features	Missing percentage (%)
Floor count	82.65%	Air temperature	0.48%
Year built	59.99%	Square feet (building size)	0.0%
Cloud coverage	43.66%	Site id	0.0%
Precip depth 1 hr	18.54%	Primary use	0.0%
Wind direction	7.17%	meter	0.0%
Sea level pressure	6.09%	timestamp	0.0%
Wind speed	0.71%	Building id	0.0%
Dew temperature	0.5%	Meter reading	0.0%

4.2. Target variable distribution

The raw meter-reading distribution was extremely right-skewed, with a skewness value of approximately 104.81. This indicates that most observations had relatively low consumption values, while a small number of observations had extremely large readings (Figure 1). Such a distribution can make regression models overly sensitive to extreme values, particularly linear models. After applying the log-plus-one transformation, the skewness decreased substantially to approximately −0.25, producing a more balanced target distribution. Therefore, the log-transformed meter reading was used as the final modelling target (see section 5.4).

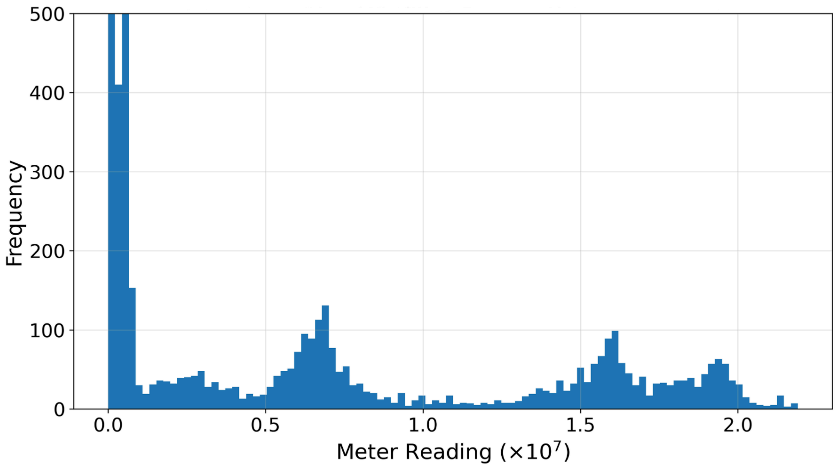

Figure 1. Raw meter reading distribution (zoomed).

4.3. Building and meter characteristics

In the four meter types, the electricity meter dominated (Figure 2), and in the primary-use category, educational institutes dominated (Figure 3). Building floor area was also right-skewed, with a raw skewness of approximately 2.67. Since building size is expected to be strongly associated with energy demand, using the raw variable could allow very large buildings to dominate the scale of the feature space. Applying a logarithmic transformation produced a more compressed representation of building size and created the ‘square feet log’ feature used in the final models (Section 5.4).

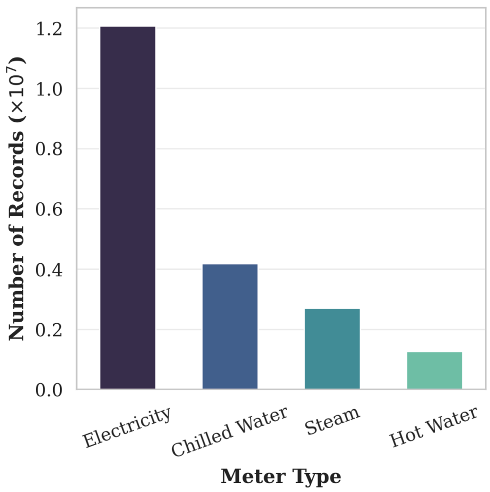

Figure 2. Distribution of meter type.

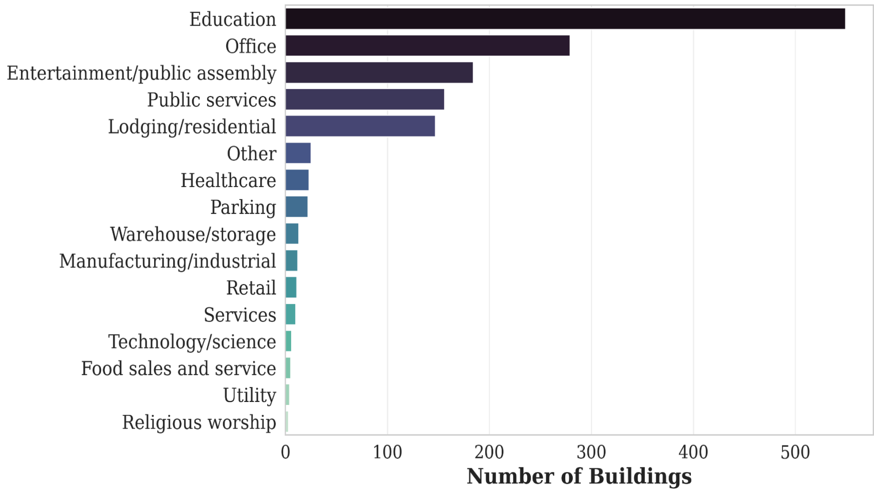

Figure 3. Distribution of primary-use categories.

4.4. Temporal consumption patterns

Figure 4 shows clear meter-specific seasonal patterns. Chilled water increased during warmer months, which is consistent with cooling demand, while steam and hot water were higher in colder months, reflecting heating-related demand. Electricity also varied across the year, although the one-year dataset means month-to-month patterns should not be interpreted as a full seasonal cycle. These trends supported the use of temporal features such as month and week of year (Section 5.5), and justified reporting meter-specific model performance later in the analysis (Section 8, Table 7).

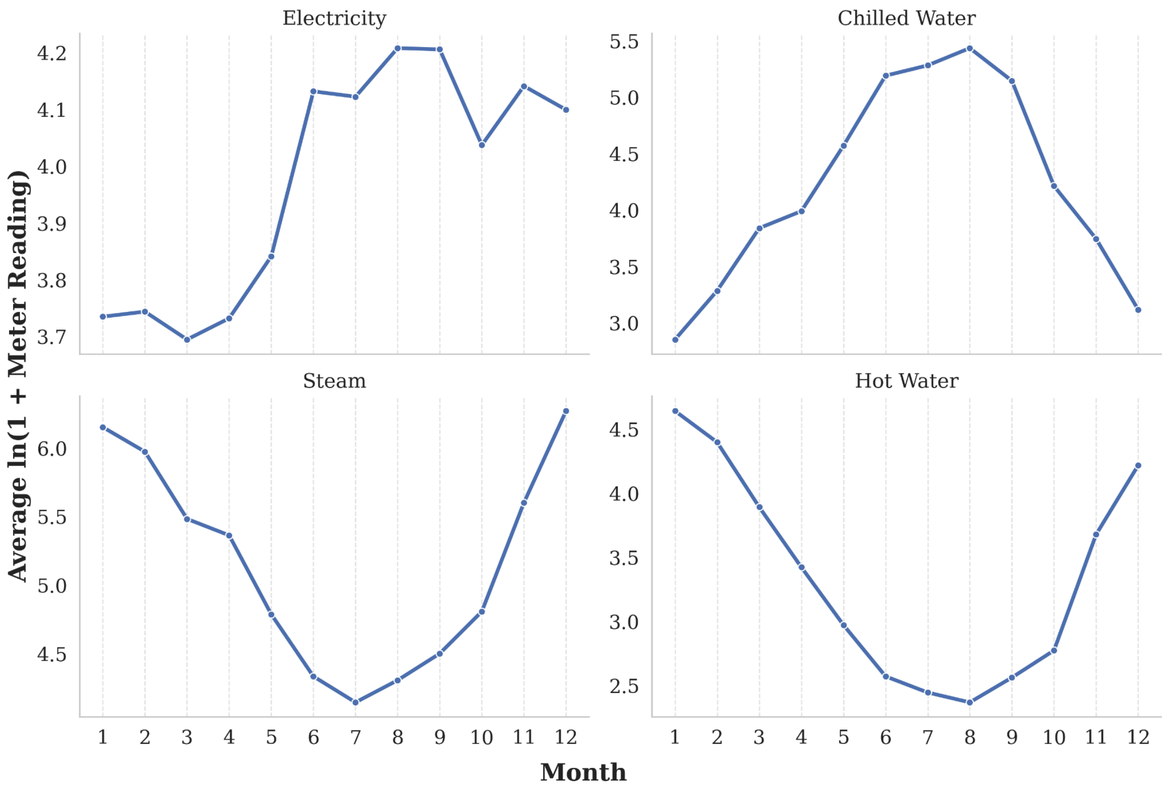

Figure 4. Average monthly log meter reading by meter type.

5. Preprocessing and Feature Engineering

Preprocessing and feature engineering were central to this project because the original dataset was not provided as a single modelling table. Instead, the prediction task required combining meter readings, building-level metadata, and site-level weather observations into a unified schema. The preprocessing pipeline was therefore designed to perform two main functions: first, to construct a clean and computationally manageable dataset; and second, to transform the raw temporal, physical, and environmental variables into features suitable for different model families. Figure 5 summarises this process, beginning with the integration of the three source files and ending with model-specific feature sets for linear and tree-based models. A key design decision was to avoid treating the task as a purely generic tabular regression problem. Since energy consumption is strongly influenced by time-dependent behaviour, occupancy patterns, weather conditions, and building characteristics, the feature engineering process explicitly preserved temporal structure through calendar-based, cyclical, and interaction features. At the same time, transformations such as logarithmic scaling were applied to reduce the effect of extreme skewness in both the target variable and building size. This resulted in a model-ready dataset that was better aligned with the underlying behaviour of building energy consumption and more appropriate for robust model comparison.

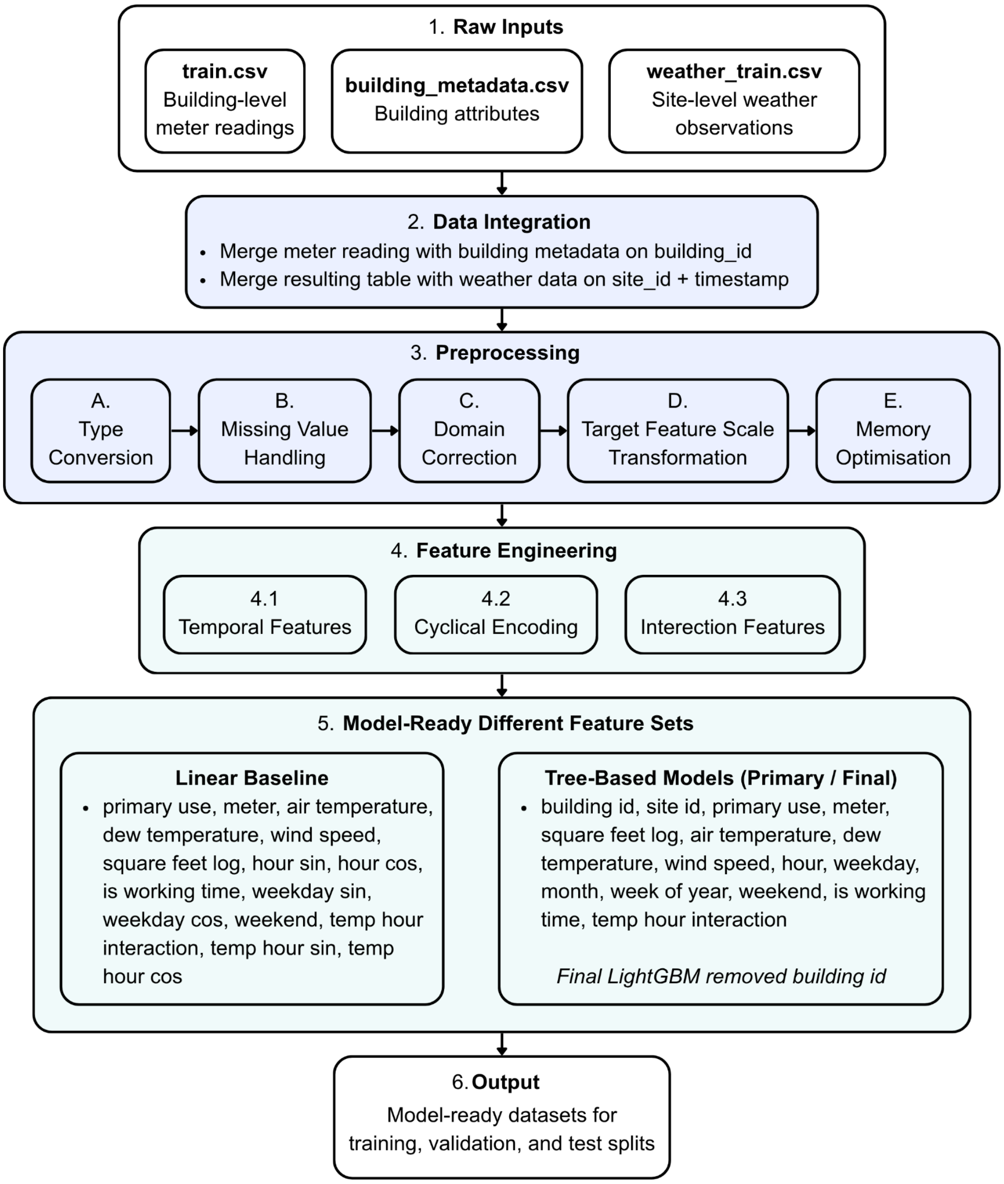

Figure 5. Preprocessing and feature engineering pipeline used to construct the final model-ready dataset. The target variable was log-transformed meter reading.

5.1. Data merging and schema creation

The three source files, train, building metadata, and weather, were merged into a unified schema: meter readings joined with building metadata on building id, then with weather observations on site id and timestamp, linking each consumption record to building characteristics and site-level weather conditions. This produced a unified building-meter-time schema in which each consumption record was linked to building characteristics, meter type, temporal information, and corresponding site-level weather conditions.

5.2. Missing-value handling and feature removal

Missing-value analysis was used to separate variables that could be reliably retained from variables that were too incomplete for robust modelling. Low-missingness continuous weather variables, specifically air temperature, dew temperature, and wind speed, were retained and imputed using the median. Median imputation was chosen because these variables may contain extreme values, and the median is less sensitive to outliers than the mean. Variables with substantial missingness (Table 2), including floor count, year built, cloud coverage, Precipitation depth over 1 hour, and sea level pressure, were removed rather than heavily imputed. This reduced the risk of introducing artificial patterns into the dataset, especially where missingness may not be completely random. Wind direction was also removed because it had limited predictive value compared with the retained weather variables. Pearson correlation was used only as a secondary exploratory heuristic, not as the sole feature-selection criterion [15]. Features were removed when weak linear association coincided with high missingness, weak domain relevance, or limited modelling value. The raw timestamp column was also removed after temporal features had been extracted from it (Section 5.5). This prevented redundancy while preserving its useful information through engineered variables such as hour, weekday, month, week of year, weekend, and working-time indicators.

5.3. Memory optimization

The merged dataset contained more than 20 million records, so memory optimisation was required before repeated model training and validation. Integer and floating-point columns were downcast to smaller numerical types where their value ranges allowed it, while care was taken not to unnecessarily reduce the precision of the target variable. This reduced the memory footprint by approximately 66%, making the workflow more scalable for model comparison and LightGBM experimentation. The processed dataset was also stored in Parquet[2] format to improve loading efficiency compared with repeatedly reading large CSV files. These steps improved computational efficiency without changing the statistical meaning of the variables.

5.4. Target correction and transformation

Before modelling, a domain-specific target correction was applied. For records where site_id = 0 and meter = 0, the meter readings were converted from kBTU to kWh using the conversion factor 0.293071. This was necessary because inconsistent measurement units can distort the target distribution and encourage the model to learn unit artefacts rather than real consumption behaviour. The conversion factor was consistent with the unit conversion process reported for the ASHRAE GEPIII Competition [11]. As identified in the EDA (Section 4.2), the raw meter-reading distribution was then examined and found to be extremely right-skewed (Figure 1). To reduce the influence of extreme consumption values, the target was transformed using:

$$y = log(1 + meter\, reading) \tag{1}$$

Where, y = log-transformed meter reading. This produced the final modelling target. The log transformation was suitable because it handles zero readings while compressing very large values. The same logic was applied to building size: square feet was also log-transformed because building floor area was also right-skewed. This produced a more balanced representation of building scale, especially for the linear models, while still remaining useful for tree-based models.

5.5. Time-aware feature engineering

The timestamp was converted into calendar-based features because building energy consumption follows daily, weekly, and seasonal patterns. The engineered temporal variables included hour, weekday, month, and week of year. Two binary indicators were also created: weekend, identifying Saturday and Sunday records, and working time, identifying readings between 8:00 and 18:00 on weekdays. These variables were designed to approximate differences in occupancy and

[2] Apache Parquet is an open-source, columnar storage file format designed for efficient data storage and retrieval.

operational behaviour across working hours, weekends, and seasonal periods. For the linear models, cyclical encodings were created for hour and weekday using sine and cosine transformations. This avoided treating adjacent cyclical values as numerically distant; for example, hour 23 and hour 0 are close in time but far apart if represented only as integers. The features hour sin, hour cos, weekday sin, and weekday cos therefore gave the linear baselines a more appropriate representation of temporal cycles. Temperature-time interaction features were also created to capture the idea that the effect of air temperature may vary by time of day. The main interaction feature was 'temp hour interaction', calculated as air temperature × hour, and was used in the tree-based models. Additional cyclical interactions, 'temp hour sin' and 'temp hour cos', were created mainly for the linear models, where explicit interaction terms are needed to represent combined effects.

Table 3. Feature engineering.

Original Feature	**Engineered Feature(s)**
Timestamp	Hour, sin(hour), cos(hour)
	Weekday, sin(weekday), cos(weekday),
	Weekend (boolean), Is working time (binary), Week of year, month
Air Temperature × Hour, Air Temperature × sin(hour), Air Temperature × cos(hour)	Temp hour interaction, Temp hour sin, Temp hour cos

5.6. Encoding strategy by model family

Different feature representations were used for linear and tree-based models because these model families handle categorical and temporal variables differently. For Linear Regression and Ridge Regression, categorical variables such as meter type and primary use were one-hot encoded, and numerical variables were standardised. Building id and site id were excluded from the linear baseline because they are high-cardinality identifiers and would substantially increase dimensionality after encoding. For Random Forest, categorical variables were also one-hot encoded because the scikit-learn implementation requires numerical input. However, feature scaling was not required because tree-based models split on feature thresholds and are not sensitive to variable scale in the same way as linear models.

For LightGBM, categorical variables were handled more directly. Variables such as building id, site id, primary use, and meter were passed as categorical features, allowing LightGBM to use them during tree construction without extensive one-hot encoding [16]. The final LightGBM experiments also compared models with and without building id. This was important because building id can be highly predictive but may also encourage building-specific memorisation. The preferred final model therefore removed building id, prioritising a more generalisable feature representation while retaining strong predictive performance.

6. Methodology and Experimental Design

The modelling task was formulated as supervised regression, with the log-transformed meter reading used as the target variable. Because the dataset contains timestamped observations, the data was split chronologically rather than randomly. Records from months 1–7 were used for training, months 8–9 for validation, and months 10–12 for final testing. After model selection, the final models were retrained on months 1–9 and evaluated on the held-out future period from months 10–12. This design preserved temporal order and reduced the risk of training on information from future periods [17]. Performance was evaluated using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2) (Equations 2-4). RMSE was treated as the primary metric because it penalises larger errors more strongly, while MAE provided a complementary average-error measure and R^2 indicated the proportion of variance explained by the model. Since the target variable was log-transformed, the reported RMSE and MAE values are errors on the log-transformed target scale rather than direct errors in raw energy units. The modelling strategy progressed from simple linear baselines to more flexible tree-based methods: Linear Regression, Ridge Regression, Random Forest, and LightGBM.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y_i})^2} \quad (2)$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \widehat{y_i}\right| \quad (3)$$

$$R^2 = 1 - \{\sum_{i=1}^{n}(y_i - \widehat{y_i})^2 \div \sum_{i=1}^{n}(y_i - \overline{y})^2\} \quad (4)$$

Where, $y_i = actual\ value$, $\widehat{y_i} = predicted\ value$, $\overline{y} = mean\ of\ actual\ values$, $n = number\ of\ observations.$

6.1. Linear Baselines: Linear Regression and Ridge Regression

Linear Regression and Ridge Regression were used as baseline models to assess how much of the target variation could be explained under a linear structure. Both models used the linear-specific feature set described in Section 5.6, including retained weather variables, log-transformed square feet, temporal indicators, cyclical encodings, temperature-time interactions, meter type, and primary use. Categorical variables were one-hot encoded and numerical variables were standardised using training-set statistics only to avoid leakage. Ridge Regression used the same feature representation as Linear Regression so that any performance difference could be attributed to L2 regularisation. Alpha values of 0.01, 0.1, 1, 10, 50, and 100 were tested on the validation set, with α = 100 selected for final evaluation. Both final linear models were retrained on months 1–9 and evaluated on months 10–12.

6.2. Random Forest

Random Forest was used as a non-linear ensemble baseline. Unlike the linear models, it can capture interactions and threshold effects without requiring them to be fully specified in advance. This made it useful for testing whether the building energy prediction task required a more flexible model structure. For Random Forest, primary use and meter were one-hot encoded using One Hot Encoder, while numerical scaling was not required because tree-based models are not sensitive to feature scale in the same way as linear models. The validation model used number of trees = 120, maximum tree depth = 18, minimum samples per leaf = 8, and maximum features per split = square root. For final testing, a stronger configuration was trained with number of trees = 200, maximum tree depth = 25, and minimum samples per leaf = 6. The final model was retrained on months 1–9 and evaluated on months 10–12.

6.3. Light Gradient-Boosting Machine (LightGBM)

The first LightGBM experiment used a baseline configuration to establish the main gradient-boosted tree benchmark. The model used a regression objective, RMSE as the validation metric, learning rate = 0.05, Maximum number of leaves = 31, feature subsampling of 0.80, row subsampling of 0.80, and early stopping on the validation set. This baseline was not intended to be the final configuration; rather, it provided a reference point for assessing whether later tuning, increased boosting rounds, and the removal of building id improved generalisation. An expanding-window validation experiment was also conducted to check whether LightGBM's performance remained stable across different future validation months. The folds used months 1–5 to validate on month 6, months 1–6 to validate on month 7, months 1–7 to validate on month 8, and months 1–8 to validate on month 9. This provided a more robust check than relying only on a single validation split.

6.3.1. LightGBM Experiments

The LightGBM experiments focused on improving predictive performance while controlling model complexity and identifier dependence. After establishing a baseline LightGBM model, the main effective tuning decision was to reduce the learning rate to 0.03 and increase the number of boosting rounds. A lower learning rate allowed the model to learn more gradually, while a larger boosting budget gave it enough iterations to improve validation performance. Since early stopping did not trigger in the initial tuned runs, the maximum number of boosting rounds was increased, with the best validation score reached at iteration 8994. Model capacity was increased using hyperparameters Maximum number of leaves = 63, while Minimum samples per leaf = 80 and L2 regularisation strength = 1.0 were used to reduce overfitting. Feature and row subsampling were also applied through feature fraction = 0.80, bagging fraction = 0.85, and Bagging frequency = 5, allowing the model to learn from different subsets of predictors and observations during training (Table 4). A more strongly regularised configuration was tested but produced weaker validation performance, suggesting that excessive constraint reduced useful model capacity. The final experiment compared LightGBM models with and without building id. Although including building id slightly improved some test metrics, the preferred final model removed it to reduce dependence on building-specific identity information. This choice prioritised a more generalisable feature representation while retaining strong predictive performance.

Table 4. Key hyperparameters of the final LightGBM model.

LightGBM Python parameter	Value	Description
learning_rate	0.03	Step size for each boosting iteration
num_leaves	63	Maximum number of leaves per tree
min_child_samples	80	Minimum samples required in a leaf
feature_fraction	0.80	Fraction of features used per iteration
bagging_fraction	0.85	Fraction of data sampled for bagging
bagging_freq	5	Frequency of bagging
lambda_l2	1.00	L2 regularisation strength
num_boost_round	8994	Final number of boosting iterations

7. Results

Table 5 reports the validation performance of the models under the chronological train-validation split. The two linear models produced very similar results, with Linear Regression and Ridge Regression both achieving a validation RMSE of approximately 1.797. This indicates that L2 regularisation had limited effect, suggesting that the main limitation was not only coefficient instability but the inability of linear models to capture the non-linear structure of building energy consumption. In contrast, Random Forest reduced validation RMSE to 1.151, showing that non-linear tree-based modelling was substantially more suitable for this task. LightGBM achieved the strongest validation performance among the main model families. The base LightGBM model obtained a validation RMSE of 0.956, and the expanding-window cross-validation experiment produced a similar mean validation RMSE of 0.950. This consistency suggests that the LightGBM approach was stable under different chronological validation settings. The final LightGBM model without the building id achieved a higher validation RMSE of 0.990 but a much lower training RMSE of 0.651. This difference shows that removing the building id made the model less dependent on direct building identity, even though this slightly reduced validation performance compared with the base model.

Table 5. Validation performance comparison.

Model	Train RMSE	Validation RMSE
Linear Regression	1.896	1.797
Ridge Regression (α = 100)	1.896	1.797
Random Forest	1.030	1.151
LightGBM (Base model)	0.710	0.956
LightGBM (Base model with CV - 4)	0.712 (mean)	0.950 (mean)
Fine-tuned LightGBM (without building id)	**0.651**	**0.990**

Table 6 presents the final test results on the held-out future period. The linear baselines performed weakest, with test R^2 values of 0.251 and 0.252. Random Forest substantially improved test performance, reducing RMSE to 1.210 and increasing R^2 to 0.666. This confirms that the prediction problem contains important non-linear relationships and interactions that are not adequately captured by linear models. The best-performing model family was LightGBM. The base LightGBM model achieved a test RMSE of 1.198, MAE of 0.696, and R^2 of 0.670. The final tuned LightGBM model without building id further improved test performance to an RMSE of 1.184, MAE of 0.690, and R^2 of 0.680. Although the LightGBM model with building id achieved a slightly lower RMSE of 1.181 and MAE of 0.641, the difference in R^2 was negligible. Therefore, the model without building id was selected as the preferred final model because it provided almost identical explanatory performance while reducing reliance on a high-cardinality identifier feature (building id). All reported errors are on the log-transformed target scale, so they should be interpreted as comparative model performance rather than direct error in raw energy units.

Table 6. Final test performance comparison.

Model	Test RMSE	Test MAE	Test R^2
Linear Regression	1.807	1.296	0.251
Ridge Regression	1.806	1.296	0.252
Random Forest	1.210	0.751	0.666
LightGBM (Base model)	1.198	0.696	0.670
Fine-tuned LightGBM (without building id)	**1.184**	**0.690**	**0.680**
Fine-tuned LightGBM (with building id)	**1.181**	**0.641**	**0.680**

8. Discussion and Error Analysis

The results show a clear progression from simple linear models to more flexible tree-based methods. Linear Regression and Ridge Regression performed similarly, indicating that regularisation alone was not enough to address the complexity of the task. Building energy consumption depends on interactions between meter type, building use, site, weather, building size, and time, which are unlikely to be fully captured by a linear additive structure. The substantial improvement from linear models to Random Forest and LightGBM therefore supports the use of non-linear ensemble methods for this dataset.

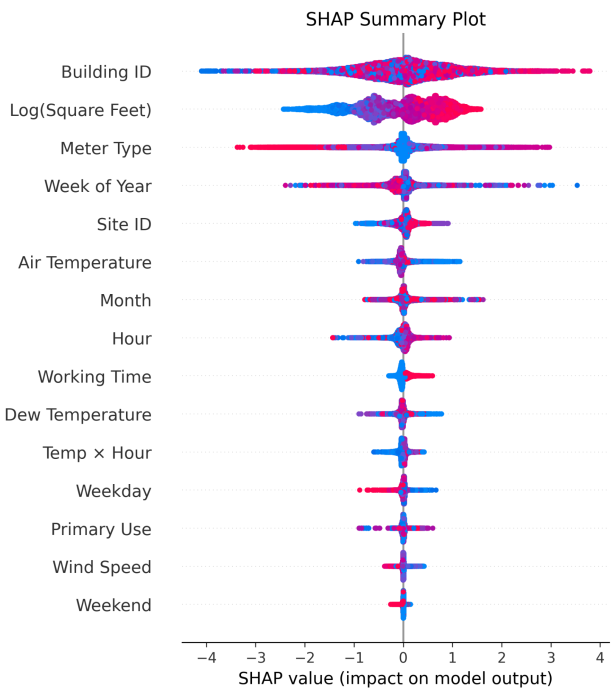

Figure 6. SHAP summary plot of LGBM with building ID.

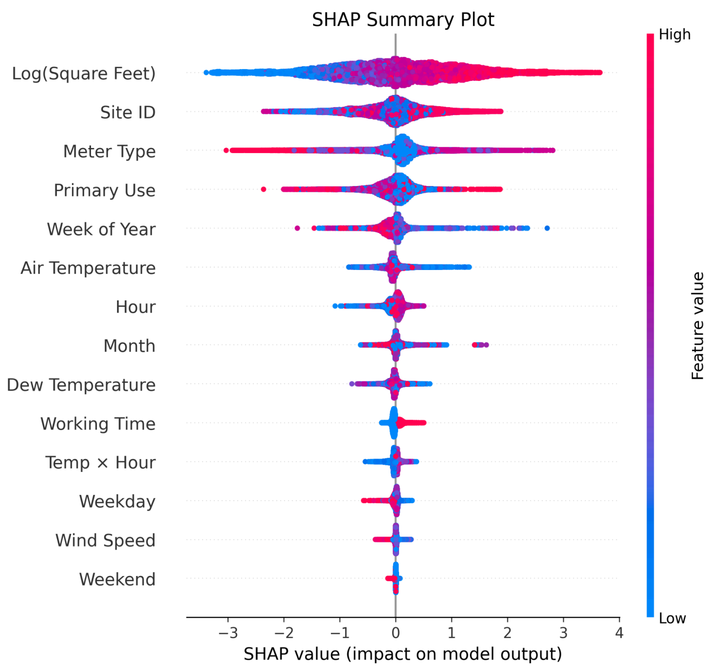

Figure 7. SHAP summary plot of LGBM without building ID.

The comparison between the two final LightGBM variants is particularly important. The model with building_id achieved slightly better RMSE and MAE, but the improvement was small. SHAP analysis showed that when building_id was included, it became one of the most influential features (Figure 6). This is not necessarily incorrect, as repeated observations from the same buildings allow the model to learn building-specific consumption patterns. However, it raises a generalisability concern: the model may partly rely on recognising known buildings rather than learning broader relationships between building characteristics, weather, time, and meter behaviour. For this reason, the model without building_id was treated as the preferred final model. It retained strong performance while shifting importance toward more general predictors such as log-transformed building size, site id, meter type, primary use, and temporal variables (Figure 7). The SHAP plots also support this interpretation. With building_id, building identity dominated the explanation, followed by building size and meter type. After removing building_id, log-transformed building size became the most important feature, which is more interpretable because larger buildings are expected to consume more energy. Site id, meter type, primary use, week of year, air temperature, hour, and month also remained influential. This pattern is consistent with the EDA, where energy consumption varied by meter type and month. Therefore, the SHAP analysis suggests that the final model relied on physically and operationally meaningful predictors rather than arbitrary identifiers alone.

The meter-wise error analysis shows that performance was not uniform across energy categories (Table 7). Electricity performed best, with the lowest RMSE of 0.786, MAE of 0.448, R^2 of 0.743, and RMSE-to-standard-deviation ratio of 0.507. This is likely due to its larger sample size and more regular usage patterns. In contrast, hot water was the weakest category, with RMSE of 1.781, MAE of 1.253, R^2 of 0.535, and the highest RMSE-to-standard-deviation ratio of 0.682. This suggests greater volatility, fewer observations, or more building-specific operational behaviour. Steam and chilled water showed intermediate performance, indicating that the model captured some seasonal heating and cooling structure but still struggled with meter-specific variability. The remaining error may reflect unavailable factors such as occupancy, equipment schedules, holidays, local operational changes, and control strategies [14].

Table 7. Final LGBM model's results and sample statistics for each meter category.

Meter type	Test samples	RMSE	MAE	R^2	Target mean	Target std.	RMSE / std.
Electricity	3,060,976	0.786	0.448	0.743	4.003	1.550	0.507
Chilledwater	1,081,665	1.583	1.002	0.607	3.690	2.525	0.627
Steam	701,454	1.536	1.012	0.616	5.554	2.478	0.620
Hotwater	319,097	1.781	1.253	0.535	3.555	2.611	0.682

Overall, the final LightGBM model achieved strong performance under the chronological test design. Since it was trained and evaluated on log-transformed meter readings, its RMSE of 1.184 is on the same logarithmic scale as the RMSLE metric used in the ASHRAE GEPIII competition. The result is competitive relative to the reported winning RMSLE of 1.231 [11], but it should not be interpreted as a direct leaderboard comparison because the official competition used a hidden future test set. The safer conclusion is that a compact, carefully engineered, single-model LightGBM pipeline produced a robust and competitive local evaluation result.

9. Limitations and Future Work

The main limitation concerns generalisation. Although building_id was removed from the preferred model to reduce direct memorisation, the chronological split still contains the same buildings in both training and testing. Therefore, the model mainly demonstrates future-period prediction for known buildings rather than true cross-building generalisation. Variables such as site id, primary use, meter type, and building size may also act as indirect building fingerprints [18]. Future work should therefore test building-level holdout splits, where entire buildings are excluded from training, and stricter site-level holdouts to evaluate generalisation to new locations and climate contexts. However, this ideas were actually restrained due to the limitation in data availability. The GEPIII dataset contains only 1,449 buildings across 15 unevenly represented sites, with one year of training records. This limits how reliably buildings or sites can be held out while still preserving enough diversity for model training and evaluation.

There are also practical and ethical considerations. Meter-wise analysis showed uneven performance, with hot water producing the weakest results. If such models were used for pricing, building ratings, or retrofit prioritisation, weaker performance for certain meter groups could create misleading or unfair decisions. Future deployment would require transparent category-specific error reporting, regular retraining, and uncertainty monitoring. Further improvements could include weather timestamp alignment, occupancy or schedule data, equipment-level information, and comparison with sequential models such as LSTM-based architectures.

10. Conclusion

This project developed a compact LightGBM pipeline for building energy prediction under the ASHRAE GEPIII framework. Tree-based models substantially outperformed the linear baselines, highlighting the importance of modelling non-linear relationships between building characteristics, meter type, weather, and temporal patterns. The preferred LightGBM model without building_id achieved an RMSE of 1.184, MAE of 0.690, and R^2 of 0.680 on the log-transformed target. Although including building_id produced marginally lower error, removing it improved interpretability and reduced reliance on building-specific identifiers. Overall, the results show that careful preprocessing, temporal feature engineering, and tuned gradient-boosted trees can deliver strong local evaluation performance with meaningful SHAP and meter-wise interpretation.

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